

Octonion Short-Time Fourier Transform for Time-Frequency Representation and Its Applications

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Abstract—The octonion Fourier transform (OFT) is a useful tool for signal processing and analysis. However, due to the lack of time localization information, it is not suitable for processing signals whose frequencies vary with time. In this paper, we utilize octonion algebra to propose a new method for time-frequency representation (TFR) called the octonion short-time Fourier transform (OSTFT). The originality of the method is based on the quaternion short-time Fourier transform (QSTFT). First, we generalize the QSTFT to the OSTFT by substituting the quaternion kernel function with the octonion kernel function in the definition of the QSTFT, and the physical significance of the OSTFT is presented. Then, several essential properties of the OSTFT are derived, such as linearity, inversion formulas, time-frequency shifts and orthogonality relations. Based on the classic Fourier convolution operation, the convolution theorem for the OSTFT is derived. We apply the relationship between the OFT and OSTFT to establish Pitt's inequality and Lieb's inequality for the OSTFT. According to the logarithmic uncertainty principle of the OFT, the logarithmic uncertainty principle associated with the OSTFT is investigated. Finally, an application in which OSTFT can be used to study linear time varying (LTV) systems is proposed, and some potential applications of the OSTFT are introduced.

Index Terms—Octonion Fourier transform, octonion short-time Fourier transform, convolution theorem, uncertainty principle, linear time-varying.

I. INTRODUCTION

THE short-time Fourier transform (STFT) is a classic and typical linear time-frequency analysis (LTFA) tool in nonstationary signal analysis. It has been used successfully for nonstationary signal separation and linear time-frequency representation (LTFR) [1]–[3]. As a special case of the STFT, the Gabor transform has been further developed [4], [5]. Based on the STFT, the synchrosqueezing transform and general linear chirplet transform were also proposed and discussed in [6]–[9]. Furthermore, over the last few years, there has been a growing interest in hypercomplex algebras, including a wide range of applications in many engineering fields [10]–[16], such as color image processing, fluid mechanics, neural networks and

signal processing. Quaternions (as the Cayley-Dickson algebra of order 4) have wide applications in signal processing [17]–[20]. The quaternion short-time Fourier transform (QSTFT), first introduced by Bahri *et al.* in [21], is particularly useful in quaternion-based signal processing and is known to be an alternative to the commonly used 2-D complex STFT. In [21], many useful properties of the QSTFT were investigated, including reconstruction formulas, reproducing kernels, isometry, and orthogonality relations. Moreover, these researchers introduced an application of the QSTFT in a linear time-varying (LTV) system. As a special case of the QSTFT, the quaternionic Gabor transform was studied in [22], [23]. Recently, some authors studied the uncertainty principles of QSTFTs [24], [25].

The octonions (as the Cayley-Dickson algebra of order 8), also known as Cayley numbers, are another hypercomplex algebra which has found many applications. In [26], based on the definition of octonion algebra and its algebraic properties, the authors derived e_4 -calculus and obtained a list of generalized Cauchy Riemann systems for octonionic monogenic functions. Hahn and Snopek [27] presented selected algebras of basis vectors for 3-D octonionic signals, and the hypothesis regarding the polar form of an octonionic signal was verified based on numerical examples. However, Bł aszczyk disproved the above result and proposed a novel numerical method in [28]. The presented numerical method in [28] gives promising results, which as a result of further work can give analytical formulas for angles in the polar representation. Snopek [29] proposed the octonion Fourier transform (OFT) of the real-valued function of three variables. The OFT is constructed by replacing the quaternion kernel function with the octonion kernel function. As a generalization of the quaternion Fourier transform (QFT), researchers have been interested in the OFT for many years. The properties of the OFT of real-valued functions of three variables and its potential applications in signal and system processing have been investigated [30], [31]. In [32], the authors derived symmetry relations, shift properties, and the Parseval and Plancherel theorems for the OFT of real-valued functions. Lian [33] exploited several uncertainty inequalities of the OFT. The properties of the OFT in the context of signal-domain operations such as derivation and convolution were also stated [34]. Recently, Bł aszczyk [12] proposed the discrete OFT, which is an excellent tool for the analysis of 3-D discrete linear time invariant (LTI) systems and 3-D discrete data.

The OFT takes functions from the time domain to the frequency domain, but it cannot be used to show the relationship between time and frequency or to perform localization for time

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and frequency simultaneously as a result of its global kernel. To overcome such a problem, the study of the Fourier transform (FT) in the octonion algebra setting motivates the interest of extending the QSTFT to octonion algebra. Therefore, in the current paper, our aim is to utilize the advantages of octonion algebra to propose a new signal transform method, which we call the octonion short-time Fourier transform (OSTFT). In some sense, the technique can be viewed as an extension of a previous work related to the QSTFT. The complex numbers are extended to higher dimensions, namely, 8, which are more flexible. The OSTFT can perform localization for time and frequency simultaneously, it can be used to observe signals and perform the analysis of time-frequency (TF) localization by a fixed TF window. Then, the physical significance of the OSTFT is given, and the properties of the OSTFT, including linearity, inversion formulas, TF shifts and orthogonality relations, are derived. Additionally, the convolution theorem for the OSTFT is explored, and Pitt's inequality and Lieb's inequality for the OSTFT are studied. According to the logarithmic uncertainty principle of the OFT, the logarithmic uncertainty principle associated with the OSTFT is investigated. Finally, an application of the OSTFT for studying LTV systems is presented, and some possible application directions are discussed.

The paper is organized as follows: the basic properties of octonion algebra are summarized and the definitions of the OFT and the QSTFT are reviewed in Section II. In Section III, the definition of the OSTFT is proposed, and the properties of the OSTFT are studied. The convolution theorem associated with the OSTFT is obtained in Section IV. Pitt's inequality and Lieb's inequality are studied in Section V. In Section VI, the logarithmic uncertainty principle of the OSTFT is derived. An application example and other potential applications of the OSTFT are presented in Section VII. In Section VIII, the conclusions of the paper are drawn.

II. PRELIMINARIES

In this section, we mainly review some basic facts regarding octonion algebra and the definitions of the OFT. Then, the QSTFT is introduced, which will be needed throughout the paper.

A. Octonion Algebra

According to the Cayley-Dickson construction [35], an arbitrary $o \in \mathbb{O}$ can be written in the form

$$o = o_0 + o_1 e_1 + o_2 e_2 + o_3 e_3 + o_4 e_4 + o_5 e_5 + o_6 e_6 + o_7 e_7,$$

where $o_0, o_1, \dots, o_7 \in \mathbb{R}$ and $e_1, e_2, \dots, e_7$ are seven different imaginary units. o_0 is called the real part of o , and the pure imaginary octonion $o_1 e_1 + o_2 e_2 + o_3 e_3 + o_4 e_4 + o_5 e_5 + o_6 e_6 + o_7 e_7$ is called the imaginary part of o . The octonion algebra is a nonassociative and noncommutative algebra. The multiplication rules in octonion algebra are presented in Table I.

The octonion conjugate is defined by

$$\bar{o} = o_0 - o_1 e_1 - o_2 e_2 - o_3 e_3 - o_4 e_4 - o_5 e_5 - o_6 e_6 - o_7 e_7.$$

TABLE I
MULTIPLICATION RULES IN OCTONION ALGEBRA

*	1	e_1	e_2	e_3	e_4	e_5	e_6	e_7
1	1	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	e_1	-1	e_3	$-e_2$	e_5	$-e_4$	$-e_7$	e_6
e_2	e_2	$-e_3$	-1	e_1	e_6	e_7	$-e_4$	$-e_5$
e_3	e_3	e_2	$-e_1$	-1	e_7	$-e_6$	e_5	$-e_4$
e_4	e_4	$-e_5$	$-e_6$	$-e_7$	-1	e_1	e_2	e_3
e_5	e_5	e_4	$-e_7$	e_6	$-e_1$	-1	$-e_3$	e_2
e_6	e_6	e_7	e_4	$-e_5$	$-e_2$	e_3	-1	$-e_1$
e_7	e_7	$-e_6$	e_5	e_4	$-e_3$	$-e_2$	e_1	-1

We can define the norm by $|o| = \sqrt{o\bar{o}} = \sqrt{\bar{o}o}$ and $|o|^2 = \sum_{j=0}^7 o_j^2$. The norm satisfies $|\epsilon\epsilon| = |\epsilon||\epsilon|$, for all $\epsilon, \epsilon \in \mathbb{O}$.

The exponential function of an octonion variable is defined by

$$e^o = \sum_{k=0}^{\infty} \frac{o^k}{k!}.$$

Every octonion $o \in \mathbb{O}$ can be represented in the form

$$o = \alpha + \beta e_4,$$

where $\alpha = o_0 + o_1 e_1 + o_2 e_2 + o_3 e_3 \in \mathbb{H}$ and $\beta = o_4 + o_5 e_1 + o_6 e_2 + o_7 e_3 \in \mathbb{H}$. This form is called the quaternionic form of an octonion.

Moreover, we have the following properties [26], [32].

Property 1: Let $\alpha, \beta \in \mathbb{H}$. Then

- (1) $e_4 \alpha = \bar{\alpha} e_4$; (2) $e_4 (\alpha e_4) = -\bar{\alpha}$; (3) $(\alpha e_4) e_4 = -\alpha$,
- (4) $\alpha (\beta e_4) = (\beta \alpha) e_4$; (5) $(\alpha e_4) \beta = (\alpha \bar{\beta}) e_4$.

Property 2: For an octonion $\alpha + \beta e_4$, $\alpha, \beta \in \mathbb{H}$ in the quaternionic form we have

$$\overline{\alpha + \beta e_4} = \bar{\alpha} - \beta e_4, \quad |\alpha + \beta e_4|^2 = |\alpha|^2 + |\beta|^2.$$

Property 3: Let $\tilde{o}, \hat{o} \in \mathbb{O}$. Then

$$e^{\tilde{o}} \cdot e^{\hat{o}} = e^{\tilde{o} + \hat{o}} \text{ if and only if } \tilde{o} \cdot \hat{o} = \hat{o} \tilde{o}.$$

For each real-valued function $f(x)$ over $\mathbb{R}^d$ and $1 \leq p < \infty$, the L^p -norm of f is defined by

$$\|f\|_{L^p(\mathbb{R}^d)} = \left(\int_{\mathbb{R}^d} |f(x)|^p dx \right)^{\frac{1}{p}},$$

where $x = (x_1, x_2, \dots, x_d) \in \mathbb{R}^d$, if $p = \infty$, then the L^∞ -norm of f is defined by

$$\|f\|_\infty = \text{ess sup}_{x \in \mathbb{R}^d} |f(x)|.$$

An inner product of real-valued functions $f(x), g(x)$ is given by

$$\langle f, g \rangle = \int_{\mathbb{R}^d} f(x) \overline{g(x)} dx.$$

Let $f, g \in L^2(\mathbb{R}^d)$, the classic convolution operation is defined as

$$(f * g)(x) = \int_{\mathbb{R}^d} f(y) g(x - y) dy.$$

B. Octonion Fourier Transform

The definition of the OFT is similar to that of the QFT. In the following part, we introduce the OFT. Unless otherwise stated, we assume that the following signal is continuous and that both the signal and its OFT are integrable (in the Lebesgue sense) in this paper. Moreover, it is necessary to note that the multiplication in the next integral is done from left to right because the octonion algebra is nonassociative.

Definition 1 (OFT [32]): The 3-D OFT of a real-valued function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ can be defined as

$$F_o(f)(w) = \int_{\mathbb{R}^3} f(x) e^{-e_1 2\pi x_1 w_1} e^{-e_2 2\pi x_2 w_2} e^{-e_4 2\pi x_3 w_3} dx, \quad (1)$$

where $x = (x_1, x_2, x_3) \in \mathbb{R}^3$, $w = (w_1, w_2, w_3) \in \mathbb{R}^3$.

In the above definition, the OFT is a special case of the n -dimensional Cayley-Dickson Fourier transform [27], and this product is an octonion-valued function. The kernel of the OFT is expanded as

$$\begin{aligned} & (e^{-e_1 2\pi x_1 w_1} \cdot e^{-e_2 2\pi x_2 w_2}) \cdot e^{-e_4 2\pi x_3 w_3} \\ &= ((c_1 - s_1 e_1) \cdot (c_2 - s_2 e_2)) \cdot (c_4 - s_4 e_4) \\ &= (c_1 c_2 - s_1 c_2 e_1 - c_1 s_2 e_2 + s_1 s_2 e_3) \cdot (c_4 - s_4 e_4) \\ &= c_1 c_2 c_4 - s_1 c_2 c_4 e_1 - c_1 s_2 c_4 e_2 + s_1 s_2 c_4 e_3 \\ &\quad - c_1 c_2 s_4 e_4 + s_1 c_2 s_4 e_5 + c_1 s_2 s_4 e_6 - s_1 s_2 s_4 e_7, \end{aligned}$$

where $c_i = \cos(2\pi x_i w_i)$, $s_i = \sin(2\pi x_i w_i)$ and $i = 1, 2, 4$. We see that this formula is a full octonion. This fact inspired the right-side QFT [27], [36], and the choice of imaginary units $\{e_1, e_2, e_4\}$ in exponents was arbitrary. Of course, other sets of imaginary units in the kernel of the Cayley-Dickson Fourier transform could be used to obtain the octonion kernel function, such as $\{e_1, e_7, e_2\}$ or $\{e_2, e_3, e_6\}$ [37]. The order of imaginary units in is strictly defined, and its change gives another definition of the OFT. However, all those definitions have the same properties, and we will only consider Definition 1.

Lemma 1: Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$. Then, the OFT of f is an invertible transform, and its inverse is given by

$$\begin{aligned} f(x) &= F_o^{-1}(F_o(f))(x) \\ &= \int_{\mathbb{R}^3} F_o(f)(w) e^{e_4 2\pi x_3 w_3} e^{e_2 2\pi x_2 w_2} e^{e_1 2\pi x_1 w_1} dw, \quad (2) \end{aligned}$$

where $x = (x_1, x_2, x_3) \in \mathbb{R}^3$, $w = (w_1, w_2, w_3) \in \mathbb{R}^3$.

The following Parseval formula of the OFT is proposed in [32].

Lemma 2 ([32]): For any real-valued functions f and g are related to their OFT via the Parseval formula, given as

$$\langle f, g \rangle = \langle F_o(f), F_o(g) \rangle. \quad (3)$$

For $f = g$, we have

$$\|f\|_{L^2(\mathbb{R}^3)} = \|F_o(f)\|_{L^2(\mathbb{R}^3)}.$$

C. Quaternionic Short-Time Fourier Transform

To derive a novel OSTFT, let us first revisit the definition of the QSTFT.

The QSTFT was introduced by Bahri *et al.*, which is a generalization of the STFT in the frame of quaternion algebra [24]. Let e_1, e_2 and e_3 (or equivalently i, j, k) denote the three imaginary units in the quaternion algebra [32].

Definition 2 (QSTFT): Let a real-valued function $f \in L^2(\mathbb{R}^2)$ and a window function $\phi \in L^2(\mathbb{R}^2) \setminus \{0\}$. Then, the QSTFT of f can be defined as

$$G_\phi f(w, u) = \int_{\mathbb{R}^2} f(x) \overline{\phi(x - u)} e^{-e_1 2\pi x_1 w_1} e^{-e_2 2\pi x_2 w_2} dx, \quad (4)$$

where $x = (x_1, x_2) \in \mathbb{R}^2$, $w = (w_1, w_2) \in \mathbb{R}^2$ and $u = (u_1, u_2) \in \mathbb{R}^2$, w and u denote frequency and time, respectively.

If the window function is the Gaussian function $\phi_\delta(u) = \frac{1}{2\delta\sqrt{\pi}} e^{-(\frac{1}{2\delta})^2 u^2}$, where $\delta = \frac{1}{\sqrt{4\pi}}$, then the QSTFT is called the quaternion Gabor transform [22], [23].

Detailed information about the QSTFT and its properties can be found in [21].

III. OCTONION SHORT-TIME FOURIER TRANSFORM

In this section, the definition and properties of the OSTFT of the real-valued function of three variables are presented.

A. Definition

Next, we propose the definition of the OSTFT.

Definition 3: Let a real-valued function $f \in L^2(\mathbb{R}^3)$ and a real-valued window function $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$. Then the OSTFT of f is defined as

$$\begin{aligned} G_o^\phi f(w, u) &= \int_{\mathbb{R}^3} f(x) \overline{\phi(x - u)} \\ &\quad \cdot e^{-e_1 2\pi x_1 w_1} e^{-e_2 2\pi x_2 w_2} e^{-e_4 2\pi x_3 w_3} dx, \quad (5) \end{aligned}$$

where $x = (x_1, x_2, x_3) \in \mathbb{R}^3$, $w = (w_1, w_2, w_3) \in \mathbb{R}^3$ and $u = (u_1, u_2, u_3) \in \mathbb{R}^3$. w and u denote frequency and time, respectively. $\bar{\phi}$ denotes the complex conjugate of ϕ because the window function ϕ is real-valued, so $\bar{\phi} = \phi$.

Formula (5) is written in terms of classic convolution

$$\begin{aligned} G_o^\phi f(w, u) &= (f(u) e^{-e_1 2\pi u_1 w_1} e^{-e_2 2\pi u_2 w_2} e^{-e_4 2\pi u_3 w_3}) \\ &\quad * \overline{\phi(-u)}, \end{aligned}$$

and the relations can be interpreted as a modulated filtering operation, as shown in Fig. 1(a). Alternatively, we can rearrange (5) as

$$\begin{aligned} G_o^\phi f(w, u) &= [(f(u) e^{-e_1 2\pi u_1 w_1} e^{-e_2 2\pi u_2 w_2}) \\ &\quad * (\overline{\phi(-u)} e^{e_4 2\pi u_3 w_3})] \cdot e^{-e_4 2\pi u_3 w_3}, \end{aligned}$$

and the relations can be interpreted as a modulated filtering operation, as shown in Fig. 1(b).

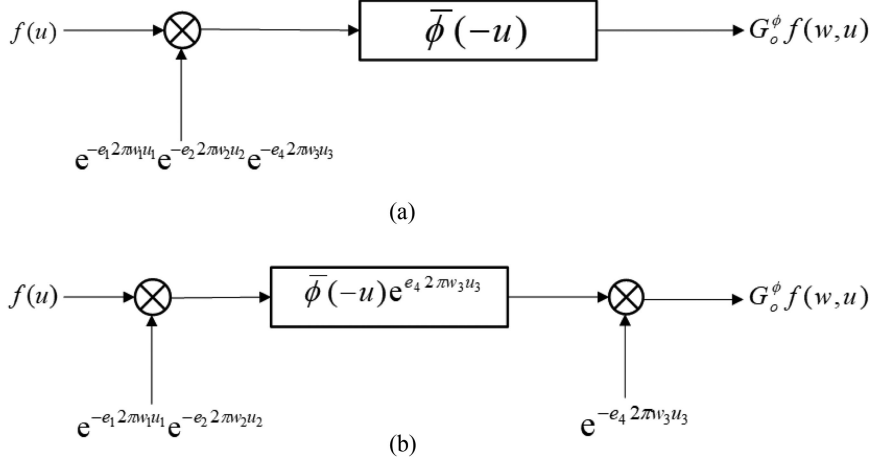


Fig. 1. Modulated filtering interpretation of OSTFT.

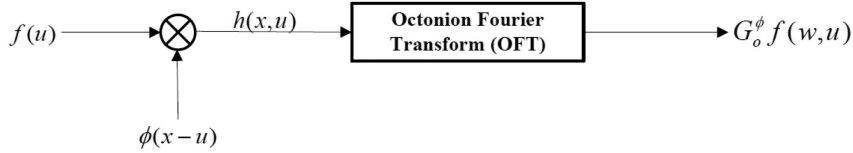


Fig. 2. Structure of computing the OSTFT.

While f is a function of time x , its OSTFT $G_o^\phi f$ is a function of time u and frequency w . Since the octonion algebra is non-associative, it should be noted that the multiplication in the above integrals is performed from left to right.

• Based on the properties of quaternion and octonion algebras, formula (5) can be rewritten in the form of an inner product as

$$G_o^\phi f(w, u) = \langle f, \Psi_{x,w,u} \rangle,$$

where $\Psi_{x,w,u} = \phi(x-u)e^{e_4 2\pi x_3 w_3} (e^{e_2 2\pi x_2 w_2} e^{e_1 2\pi x_1 w_1})$ is the kernel of the OSTFT.

• Integrating the OSTFT with respect to the time variable yields the OFT multiplied by a constant

$$\int_{\mathbb{R}^3} G_o^\phi f(w, u) du = \Omega \cdot F_o(f)(w),$$

where the coefficient $\Omega = \int_{\mathbb{R}^3} \bar{\phi}(t) dt$.

• The definition of the OSTFT can be expressed as

$$G_o^\phi f(w, u) = F_o(h)(w). \quad (6)$$

where $h(x, u) = f(x)\overline{\phi(x-u)}$, $F_o(w)$ denote the OFT of $h(x, u)$. Formula (6) consists of the following two steps:

i) Multiplying a signal by a window function, i.e.,

$$f(x) \Rightarrow h(x, u) = f(x)\phi(x-u);$$

ii) The OFT of $h(x, u)$, i.e.,

$$h(x, u) \Rightarrow F_o(h)(w) \Rightarrow G_o^\phi f(w, u).$$

This implies that all results for the OFT can be easily extended to the proposed OSTFT, and vice versa as shown in Fig. 2. This is consistent with the relationship between some other existing

transforms and their conventional counterparts. Furthermore, the computational complexity of the proposed OSTFT depends on its OFT counterpart.

• Time-frequency localization: According to the form of an inner product of the OSTFT, if the novel OSTFT kernel $\Psi_{x,w,u}$ has bounded support in the time domain, then the inner product of $f(x)$ and $\Psi_{x,w,u}(x)$ also has bounded support in the time domain. Hence, $G_o^\phi(f)(w, u)$ contains information about $f(x)$ near u . In addition, by (6), $F_o(h)(w)$ can provide frequency-domain local properties. This implies that the novel OSTFT is capable of providing time- and frequency-domain information simultaneously.

• The energy density of the OSTFT is defined by

$$|G_o^\phi f(w, u)|^2 = \left| \int_{\mathbb{R}^3} f(x) \overline{\phi(x-u)} e^{-e_1 2\pi x_1 w_1} e^{-e_2 2\pi x_2 w_2} e^{-e_4 2\pi x_3 w_3} dx \right|^2,$$

which measures the energy of a signal in the TF plane in the neighborhood of point (w, u) .

Lemma 3: Let $\phi \in L^p(\mathbb{R}^3)$, $f \in L^q(\mathbb{R}^3)$, $p, q \in [1, \infty]$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then

$$|G_o^\phi f(w, u)| \leq \|\phi\|_{L^p(\mathbb{R}^3)} \|f\|_{L^q(\mathbb{R}^3)}.$$

Proof: Applying Holder's inequality, we can obtain the result.

The following result describes an inequality related to the OSTFT.

Lemma 4: Let $\phi \in L^p(\mathbb{R}^3)$ and $f \in L^1(\mathbb{R}^3)$. Then

$$\|G_o^\phi f(w, \cdot)\|_{L^p(\mathbb{R}^3)} \leq \|\phi\|_{L^p(\mathbb{R}^3)} \|f\|_{L^1(\mathbb{R}^3)}.$$

Proof: According to Minkowski's inequality, we have

$$\begin{aligned} \|G_o^\phi f(w, \cdot)\|_{L^p(\mathbb{R}^3)} &\leq \int_{\mathbb{R}^3} \left(\int_{\mathbb{R}^3} \left| f(x) \overline{\phi(x-u)} e^{-e_1 2\pi x_1 w_1} \right. \right. \\ &\quad \left. \left. \cdot e^{-e_2 2\pi x_2 w_2} e^{-e_4 2\pi x_3 w_3} \right|^p du \right)^{\frac{1}{p}} dx \\ &= \int_{\mathbb{R}^3} \left(\int_{\mathbb{R}^3} \left| f(x) \overline{\phi(z)} \right|^p dz \right)^{\frac{1}{p}} dx. \end{aligned}$$

Substituting $x - u = z$ in the above inequality, we obtain

$$\begin{aligned} \|G_o^\phi f(w, \cdot)\|_{L^p(\mathbb{R}^3)} &\leq \int_{\mathbb{R}^3} \left(\int_{\mathbb{R}^3} \left| f(x) \overline{\phi(z)} \right|^p dz \right)^{\frac{1}{p}} dx \\ &= \left(\int_{\mathbb{R}^3} \left| \overline{\phi(z)} \right|^p dz \right)^{\frac{1}{p}} \int_{\mathbb{R}^3} |f(x)| dx, \end{aligned}$$

we obtain the result.

B. Properties

In [34], some properties of the OFT are obtained. In this subsection, we discuss several basic properties of the OSTFT.

Theorem 1: Let real-valued functions $f, g \in L^2(\mathbb{R}^3)$ and a real-valued window function $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$. For any $a, b \in \mathbb{R}$, then the OSTFT is a linear operator, namely,

$$G_o^\phi(af + bg) = aG_o^\phi(f) + bG_o^\phi(g). \quad (7)$$

Proof: This discussion follows directly from the linearity of the product and the integration involved in Definition 3.

The theorem implies that the novel OSTFT has no cross-term interference and is suitable for analyzing multi-component signals.

Theorem 2: Let $f \in L^2(\mathbb{R}^3)$ be a real-valued function, two real-valued time-domain window functions $\phi, \varphi \in L^2(\mathbb{R}^3) \setminus \{0\}$ and $\langle \phi, \varphi \rangle \neq 0$. Then we have the inversion formula of the OSTFT

$$\begin{aligned} f(x) &= \frac{1}{\langle \phi, \varphi \rangle} \int_{\mathbb{R}^6} G_o^\phi f(w, u) e^{e_4 2\pi x_3 w_3} \\ &\quad \cdot e^{e_2 2\pi x_2 w_2} e^{e_1 2\pi x_1 w_1} \varphi(x-u) dw du. \end{aligned} \quad (8)$$

Proof: Let $h(x, u) = f(x) \overline{\phi(x-u)}$, according to Lemma 1, we obtain the inversion formula of the OFT of the function h .

$$\begin{aligned} h(x, u) &= F_o^{-1}(F_o(h))(x) \\ &= \int_{\mathbb{R}^3} F_o(h)(w) e^{e_4 2\pi x_3 w_3} e^{e_2 2\pi x_2 w_2} e^{e_1 2\pi x_1 w_1} dw. \end{aligned} \quad (9)$$

Hence, from (6), we have

$$\begin{aligned} f(x) \overline{\phi(x-u)} &= \int_{\mathbb{R}^3} G_o^\phi f(w, u) e^{e_4 2\pi x_3 w_3} e^{e_2 2\pi x_2 w_2} e^{e_1 2\pi x_1 w_1} dw. \end{aligned} \quad (10)$$

Multiplying both sides of (10) from the right by $\varphi(x-u)$, we have

$$f(x) \overline{\phi(x-u)} \varphi(x-u)$$

$$= \int_{\mathbb{R}^3} G_o^\phi f(w, u) e^{e_4 2\pi x_3 w_3} e^{e_2 2\pi x_2 w_2} e^{e_1 2\pi x_1 w_1} \varphi(x-u) dw.$$

Then integrating with respect to du in the above formula, we obtain

$$\begin{aligned} &\int_{\mathbb{R}^3} f(x) \overline{\phi(x-u)} \varphi(x-u) du \\ &= \int_{\mathbb{R}^6} G_o^\phi f(w, u) e^{e_4 2\pi x_3 w_3} e^{e_2 2\pi x_2 w_2} e^{e_1 2\pi x_1 w_1} \varphi(x-u) dw du. \end{aligned}$$

Hence

$$\begin{aligned} f(x) \langle \phi, \varphi \rangle &= \int_{\mathbb{R}^6} G_o^\phi f(w, u) e^{e_4 2\pi x_3 w_3} e^{e_2 2\pi x_2 w_2} e^{e_1 2\pi x_1 w_1} \varphi(x-u) dw du. \end{aligned}$$

The inversion formula (8) shows that f can be expressed as a continuous superposition of TF shifts with the OSTFT as a weight function.

Assuming that

$$\begin{aligned} \Phi(x) &= \int_{\mathbb{R}^6} G_o^\phi f(w, u) e^{e_4 2\pi x_3 w_3} e^{e_2 2\pi x_2 w_2} e^{e_1 2\pi x_1 w_1} \\ &\quad \cdot \varphi(x-u) dw du, \end{aligned}$$

then $f(x) = \frac{1}{\langle \phi, \varphi \rangle} \Phi(x)$, which indicates that if window functions $\phi(x)$ and $\varphi(x)$ satisfy

$$\langle \phi, \varphi \rangle = 1, \quad (11)$$

then we obtain $f(x) = \Phi(x)$. (11) is called the perfect reconstruction condition of the novel OSTFT. Mathematically, for a given window function $\phi(x)$, there exists an infinite number of choices for the window function $\varphi(x)$ that satisfy (11). Here, we illustrate two simple options: (1) $\phi(x) = \varphi(x)$ and (2) $\varphi(x) = 1$. We consider the first option, i.e., $\phi(x) = \varphi(x)$. Then, the perfect reconstruction condition (11) becomes $\|\phi\|_{L^2(\mathbb{R}^3)}^2 = 1$. In this case, the inverse OSTFT is written as

$$\begin{aligned} f(x) &= \int_{\mathbb{R}^6} G_o^\phi f(w, u) e^{e_4 2\pi x_3 w_3} e^{e_2 2\pi x_2 w_2} e^{e_1 2\pi x_1 w_1} \\ &\quad \cdot \phi(x-u) dw du. \end{aligned}$$

Next, we will show that the OSTFT is related to the 3-D STFT. This proof is identical to that in [34].

Theorem 3: Let $f \in L^2(\mathbb{R}^3)$ be a real-valued function, $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a real-valued window function and $V_\phi f(w, u)$ be called the 3-D STFT of a function f with respect to ϕ . Then, we have the relation between the OSTFT and the 3-D STFT :

$$\begin{aligned} G_o^\phi f(w, u) &= \frac{1}{4} \{ (V_\phi f(w_1, w_2, w_3; u) + V_\phi f(w_1, w_2, -w_3; u))(1 - e_3) \\ &\quad + (V_\phi f(w_1, -w_2, w_3; u) + V_\phi f(w_1, -w_2, -w_3; u))(1 + e_3) \} \\ &\quad + \frac{1}{4} \{ (V_\phi f(w_1, w_2, -w_3; u) - V_\phi f(w_1, w_2, w_3; u))(1 - e_3) \\ &\quad + (V_\phi f(w_1, -w_2, -w_3; u) - V_\phi f(w_1, -w_2, w_3; u))(1 + e_3) \} \\ &\quad \cdot e_5. \end{aligned} \quad (12)$$

Proof: Let us recall the definition of the STFT given by [1]

$$V_\phi f(w, u) = \int_{\mathbb{R}^3} f(x) \overline{\phi(x-u)} e^{-e_1 \alpha_1} e^{-e_1 \alpha_2} e^{-e_1 \alpha_3} dx, \quad (13)$$

where $\alpha_1 = 2\pi x_1 w_1$, $\alpha_2 = 2\pi x_2 w_2$ and $\alpha_3 = 2\pi x_3 w_3$.

According to the equivalent definition of sine and cosine functions, we obtain

$$\begin{aligned} & \frac{1}{2} (V_\phi f(w_1, w_2, w_3; u) + V_\phi f(w_1, -w_2, w_3; u)) \\ &= \int_{\mathbb{R}^3} f(x) \overline{\phi(x-u)} e^{-e_1 \alpha_1} (\cos \alpha_2) e^{-e_1 \alpha_3} dx, \end{aligned} \quad (14)$$

$$\begin{aligned} & \frac{1}{2} (V_\phi f(w_1, w_2, w_3; u) - V_\phi f(w_1, -w_2, w_3; u)) \\ &= \int_{\mathbb{R}^3} f(x) \overline{\phi(x-u)} e^{-e_1 \alpha_1} (-e_1 \sin \alpha_2) e^{-e_1 \alpha_3} dx. \end{aligned} \quad (15)$$

By changing the sign of w_3 in (15), multiplying (15) from the right by e_3 and applying the multiplication rules from Table I, we have

$$\begin{aligned} & \frac{1}{2} (V_\phi f(w_1, w_2, -w_3; u) - V_\phi f(w_1, -w_2, -w_3; u)) e_3 \\ &= \int_{\mathbb{R}^3} f(x) \overline{\phi(x-u)} e^{-e_1 \alpha_1} (e_2 \sin \alpha_2) e^{-e_1 \alpha_3} dx. \end{aligned} \quad (16)$$

By changing the sign of w_2 in (16), we have

$$\begin{aligned} & \frac{1}{2} (V_\phi f(w_1, -w_2, -w_3; u) - V_\phi f(w_1, w_2, -w_3; u)) e_3 \\ &= \int_{\mathbb{R}^3} f(x) \overline{\phi(x-u)} e^{-e_1 \alpha_1} (-e_2 \sin \alpha_2) e^{-e_1 \alpha_3} dx. \end{aligned} \quad (17)$$

Adding (14) and (17), we obtain

$$\begin{aligned} & \frac{1}{2} (V_\phi f(w_1, w_2, w_3; u) + V_\phi f(w_1, -w_2, w_3; u)) \\ &+ \frac{1}{2} (V_\phi f(w_1, -w_2, -w_3; u) - V_\phi f(w_1, w_2, -w_3; u)) e_3 \\ &= \int_{\mathbb{R}^3} f(x) \overline{\phi(x-u)} e^{-e_1 \alpha_1} e^{-e_2 \alpha_2} e^{-e_1 \alpha_3} dx. \end{aligned} \quad (18)$$

To simplify the notation, let

$$\begin{aligned} G(w, u) &= \frac{1}{2} (V_\phi f(w_1, w_2, w_3; u) + V_\phi f(w_1, -w_2, w_3; u)) \\ &+ \frac{1}{2} (V_\phi f(w_1, -w_2, -w_3; u) \\ &- V_\phi f(w_1, w_2, -w_3; u)) e_3. \end{aligned} \quad (19)$$

According to the same method, we have

$$\begin{aligned} & \frac{1}{2} (G(w_1, w_2, w_3; u) + G(w_1, w_2, -w_3; u)) \\ &= \int_{\mathbb{R}^3} f(x) \overline{\phi(x-u)} e^{-e_1 \alpha_1} e^{-e_2 \alpha_2} (\cos \alpha_3) dx, \end{aligned} \quad (20)$$

$$\begin{aligned} & \frac{1}{2} (G(w_1, -w_2, w_3; u) - G(w_1, -w_2, -w_3; u)) \\ &= \int_{\mathbb{R}^3} f(x) \overline{\phi(x-u)} e^{-e_1 \alpha_1} e^{e_2 \alpha_2} (-e_1 \sin \alpha_3) dx. \end{aligned} \quad (21)$$

Multiplying (21) from the right by $-e_5$ and applying the multiplication rules from Table I, we have

$$\begin{aligned} & \frac{1}{2} (G(w_1, -w_2, -w_3; u) - G(w_1, -w_2, w_3; u)) e_5 \\ &= \int_{\mathbb{R}^3} f(x) \overline{\phi(x-u)} e^{-e_1 \alpha_1} e^{-e_2 \alpha_2} (-e_4 \sin \alpha_3) dx. \end{aligned} \quad (22)$$

Adding (20) and (22), we obtain

$$\begin{aligned} & \frac{1}{2} (G(w_1, w_2, w_3; u) + G(w_1, w_2, -w_3; u)) \\ &+ \frac{1}{2} (G(w_1, -w_2, -w_3; u) - G(w_1, -w_2, w_3; u)) e_5 \\ &= \int_{\mathbb{R}^3} f(x) \overline{\phi(x-u)} e^{-e_1 \alpha_1} e^{-e_2 \alpha_2} e^{-e_4 \alpha_3} dx. \end{aligned} \quad (23)$$

Finally, substituting (19) in (23), we obtain the result.

Theorem 4 (Shift): Let $f \in L^2(\mathbb{R}^3)$ be a real-valued function and $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a real-valued window function. Assume that $a, b, c, a', b', c' \in \mathbb{R}$ and $T_a f(x) = f(x_1 - a, x_2, x_3)$, $T_{a'} \phi(x) = \phi(x_1 - a', x_2, x_3)$, $T_b f(x) = f(x_1, x_2 - b, x_3)$, $T_{b'} \phi(x) = \phi(x_1, x_2 - b', x_3)$, $T_c f(x) = f(x_1, x_2, x_3 - c)$ and $T_{c'} \phi(x) = \phi(x_1, x_2, x_3 - c')$. Then, we can obtain

$$\begin{aligned} G_o^{(T_{a'} \phi)}(T_a f)(w, u) &= \cos(2\pi w_1 a) G_o^\phi f(w, z_a) \\ &- \sin(2\pi w_1 a) G_o^\phi f(w_a, z_a) \cdot e_1, \end{aligned} \quad (24)$$

$$\begin{aligned} G_o^{(T_{b'} \phi)}(T_b f)(w, u) &= \cos(2\pi w_2 b) G_o^\phi f(w, z_b) \\ &- \sin(2\pi w_2 b) G_o^\phi f(w_b, z_b) \cdot e_2, \end{aligned} \quad (25)$$

$$\begin{aligned} G_o^{(T_{c'} \phi)}(T_c f)(w, u) &= \cos(2\pi w_3 c) G_o^\phi f(w, z_c) \\ &- \sin(2\pi w_3 c) G_o^\phi f(w, z_c) \cdot e_4, \end{aligned} \quad (26)$$

where $z_a = (a' + (u_1 - a), u_2, u_3)$, $z_b = (u_1, b' + (u_2 - b), u_3)$, $z_c = (u_1, u_2, c' + (u_3 - c))$, $w_a = (w_1, -w_2, -w_3)$, $w_b = (w_1, w_2, -w_3)$.

The theorem implies that a time shift in the signal causes the same time shift and frequency reversal in the OSTFT. If the window function is a unit window function, the Theorem 4 becomes the shift property of the OFT [34].

Theorem 5: Let $f \in L^2(\mathbb{R}^3)$ be a real-valued function and $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a real-valued window function. Assume that for all $w_0, w' \in \mathbb{R}$, $\zeta_i = 2\pi w_0 x_i$, $\zeta'_i = 2\pi w' x_i$ ($i = 1, 2, 3$), $M_{s,1}^{w_0} f(x) = f(x) \sin \zeta_1$, $M_{s,1}^{w'} \phi(x) = \phi(x) \sin \zeta'_1$, $M_{s,2}^{w_0} f(x) = f(x) \sin \zeta_2$, $M_{s,2}^{w'} \phi(x) = \phi(x) \sin \zeta'_2$, $M_{s,3}^{w_0} f(x) = f(x) \sin \zeta_3$ and $M_{s,3}^{w'} \phi(x) = \phi(x) \sin \zeta'_3$. Then, we can obtain (27)–(29), which appear at the bottom of the next page.

Theorem 6: Let $f \in L^2(\mathbb{R}^3)$ be a real-valued function and $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a real-valued window function. Assume that for all $w_0, w' \in \mathbb{R}$, $\zeta_i = 2\pi w_0 x_i$, $\zeta'_i = 2\pi w' x_i$ ($i = 1, 2, 3$), $M_{c,1}^{w_0} f(x) = f(x) \cos \zeta_1$, $M_{c,1}^{w'} \phi(x) = \phi(x) \cos \zeta'_1$, $M_{c,2}^{w_0} f(x) = f(x) \cos \zeta_2$, $M_{c,2}^{w'} \phi(x) = \phi(x) \cos \zeta'_2$, $M_{c,3}^{w_0} f(x) = f(x) \cos \zeta_3$ and $M_{c,3}^{w'} \phi(x) = \phi(x) \cos \zeta'_3$. Then, we can obtain (30)–(32), shown at the bottom of the next page.

Theorem 7: Let $f \in L^2(\mathbb{R}^3)$ be a real-valued function and $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a real-valued window function.

Assume that $w_0, w' \in \mathbb{R}$ and $M_{e,1}^{w_0} f(x) = f(x)e^{-e_1 2\pi w_0 x_1}$, $M_{e,1}^{w'} \phi(x) = \phi(x)e^{-e_1 2\pi w' x_1}$, $M_{e,2}^{w_0} f(x) = f(x)e^{-e_2 2\pi w_0 x_2}$, $M_{e,2}^{w'} \phi(x) = \phi(x)e^{-e_2 2\pi w' x_2}$, $M_{e,3}^{w_0} f(x) = f(x)e^{-e_4 2\pi w_0 x_3}$ and $M_{e,3}^{w'} \phi(x) = \phi(x)e^{-e_4 2\pi w' x_3}$. Then, we can obtain (33)–(35), shown at the bottom of the next page.

The theorem states that a frequency shift in the signal induces a frequency shift but no time shift in the OSTFT.

The proofs of the above theorems will be omitted, and they are similar to the proof process in [34].

Based on Theorem 4 and Theorem 7, we can obtain the theorem of TF shifts.

Theorem 8 (Time-frequency shifts): Let $f \in L^2(\mathbb{R}^3)$ be a real-valued function and $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a real-valued window function. Operators of the form $M_{e,*}^* T_*$ is called time-frequency shifts. Then, we can obtain (36)–(38), shown at the bottom of the next page.

Proof: According to the Theorem 4 and Theorem 7, we can obtain the theorem.

Theorem 9 (Orthogonality relation): Let $f, g \in L^2(\mathbb{R}^3)$ be two real-valued functions and $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a real-valued window function. Then we have

$$\begin{aligned} & \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} G_o^\phi f(w, u) \overline{G_o^\phi g(w, u)} dw du \\ &= \|\phi\|_{L^2(\mathbb{R}^3)}^2 \int_{\mathbb{R}^3} f(x) \overline{g(x)} dx. \end{aligned} \quad (39)$$

Proof: Let $h(x, u) = f(x) \overline{\phi(x - u)}$, $l(x, u) = g(x) \overline{\phi(x - u)}$. According to Lemma 2, we have

$$\int_{\mathbb{R}^3} F_o(h)(w) \overline{F_o(l)(w)} dw = \int_{\mathbb{R}^3} h(x, u) \overline{l(x, u)} dx, \quad (40)$$

$$\begin{aligned} G_o^{(M_{s,1}^{w'} \phi)}(M_{s,1}^{w_0} f)(w, u) &= \frac{1}{4} \cos(2\pi u_1 w') (G_o^\phi f(w_{s,3}^1, u) + G_o^\phi f(w_{s,5}^1, u) - G_o^\phi f(w_{s,1}^1, u) - G_o^\phi f(w_{s,7}^1, u)) \\ &\quad + \frac{1}{4} \sin(2\pi u_1 w') (G_o^\phi f(w_{s,4}^1, u) + G_o^\phi f(w_{s,8}^1, u) - G_o^\phi f(w_{s,2}^1, u) - G_o^\phi f(w_{s,6}^1, u)) e_1, \end{aligned} \quad (27)$$

$$\begin{aligned} G_o^{(M_{s,2}^{w'} \phi)}(M_{s,2}^{w_0} f)(w, u) &= \frac{1}{4} \cos(2\pi u_2 w') (G_o^\phi f(w_{s,3}^2, u) + G_o^\phi f(w_{s,5}^2, u) - G_o^\phi f(w_{s,1}^2, u) - G_o^\phi f(w_{s,7}^2, u)) \\ &\quad + \frac{1}{4} \sin(2\pi u_2 w') (G_o^\phi f(w_{s,4}^2, u) + G_o^\phi f(w_{s,8}^2, u) - G_o^\phi f(w_{s,2}^2, u) - G_o^\phi f(w_{s,6}^2, u)) e_2, \end{aligned} \quad (28)$$

$$G_o^{(M_{s,3}^{w'} \phi)}(M_{s,3}^{w_0} f)(w, u) = \frac{1}{4} (G_o^\phi f(w_{s,1}^3, u) - G_o^\phi f(w_{s,2}^3, u)) e^{e_4 2\pi u_1 w'} + \frac{1}{4} (G_o^\phi f(w_{s,3}^3, u) - G_o^\phi f(w_{s,4}^3, u)) e^{-e_4 2\pi u_1 w'}, \quad (29)$$

where $w_{s,1}^1 = (w_1 + w_0 + w', w_2, w_3)$, $w_{s,2}^1 = (w_1 + w_0 + w', -w_2, -w_3)$, $w_{s,3}^1 = (w_1 + w_0 - w', w_2, w_3)$, $w_{s,4}^1 = (w_1 + w_0 - w', -w_2, -w_3)$, $w_{s,5}^1 = (w_1 - w_0 + w', w_2, w_3)$, $w_{s,6}^1 = (w_1 - w_0 + w', -w_2, -w_3)$, $w_{s,7}^1 = (w_1 - w_0 - w', w_2, w_3)$, $w_{s,8}^1 = (w_1 - w_0 - w', -w_2, -w_3)$, $w_{s,1}^2 = (w_1, w_2 + w_0 + w', w_3)$, $w_{s,2}^2 = (w_1, w_2 + w_0 + w', -w_3)$, $w_{s,3}^2 = (w_1, w_2 + w_0 - w', w_3)$, $w_{s,4}^2 = (w_1, w_2 + w_0 - w', -w_3)$, $w_{s,5}^2 = (w_1, w_2 - w_0 + w', w_3)$, $w_{s,6}^2 = (w_1, w_2 - w_0 + w', -w_3)$, $w_{s,7}^2 = (w_1, w_2 - w_0 - w', w_3)$, $w_{s,8}^2 = (w_1, w_2 - w_0 - w', -w_3)$, $w_{s,1}^3 = (w_1, w_2, w_3 - w_0 + w')$, $w_{s,2}^3 = (w_1, w_2, w_3 + w_0 + w')$, $w_{s,3}^3 = (w_1, w_2, w_3 + w_0 - w')$, $w_{s,4}^3 = (w_1, w_2, w_3 - w_0 - w')$.

$$G_o^{(M_{c,1}^{w'} \phi)}(M_{c,1}^{w_0} f)(w, u) = \frac{1}{4} e^{-e_1 2\pi u_1 w'} (G_o^\phi f(w_{s,7}^1, u) + G_o^\phi f(w_{s,3}^1, u)) + \frac{1}{4} e^{e_1 2\pi u_1 w'} (G_o^\phi f(w_{s,5}^1, u) + G_o^\phi f(w_{s,1}^1, u)), \quad (30)$$

$$\begin{aligned} G_o^{(M_{c,2}^{w'} \phi)}(M_{c,2}^{w_0} f)(w, u) &= \frac{1}{4} \cos(2\pi u_2 w') (G_o^\phi f(w_{s,1}^2, u) + G_o^\phi f(w_{s,3}^2, u) + G_o^\phi f(w_{s,5}^2, u) + G_o^\phi f(w_{s,7}^2, u)) \\ &\quad + \frac{1}{4} \sin(2\pi u_2 w') (G_o^\phi f(w_{s,2}^2, u) + G_o^\phi f(w_{s,6}^2, u) - G_o^\phi f(w_{s,4}^2, u) - G_o^\phi f(w_{s,8}^2, u)) e_2, \end{aligned} \quad (31)$$

$$G_o^{(M_{c,3}^{w'} \phi)}(M_{c,3}^{w_0} f)(w, u) = \frac{1}{4} (G_o^\phi f(w_{s,1}^3, u) + G_o^\phi f(w_{s,2}^3, u)) e^{e_4 2\pi u_1 w'} + \frac{1}{4} (G_o^\phi f(w_{s,3}^3, u) + G_o^\phi f(w_{s,4}^3, u)) e^{-e_4 2\pi u_1 w'}. \quad (32)$$

where $F_o(h)$ and $F_o(l)$ are the OFT of h and l , respectively. Integrating both sides with respect to du in (40). We obtain

$$\begin{aligned} & \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} F_o(h)(w) \overline{F_o(l)(w)} dw du \\ &= \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} h(x, u) \overline{l(x, u)} dx du \\ &= \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} f(x) \overline{\phi(x-u)} \phi(x-u) \overline{g(x)} dx du \\ &= \|\phi\|_{L^2(\mathbb{R}^3)}^2 \int_{\mathbb{R}^3} f(x) \overline{g(x)} dx. \end{aligned} \quad (41)$$

By (6), we obtain

$$\begin{aligned} & \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} G_o^\phi f(w, u) \overline{G_o^\phi g(w, u)} dw du \\ &= \|\phi\|_{L^2(\mathbb{R}^3)}^2 \int_{\mathbb{R}^3} f(x) \overline{g(x)} dx. \end{aligned} \quad (42)$$

Based on the above theorem, we can conclude the following important consequence.

Corollary 1: If $f = g$, then

$$\int_{\mathbb{R}^3} \int_{\mathbb{R}^3} |G_o^\phi f(w, u)|^2 dw du = \|\phi\|_{L^2(\mathbb{R}^3)}^2 \|f\|_{L^2(\mathbb{R}^3)}^2. \quad (43)$$

IV. CONVOLUTION THEOREM

In this section, based on the classic Fourier convolution operation, we derive the convolution theorem for the OSTFT. Many papers have introduced that the OFT of the real-valued function can be expressed as the octonion sum of components of different parities [27], [32], [36]. According to (6), the OSTFT of the real-valued function can also be expressed as the octonion sum of components of different parities, i.e.

$$\begin{aligned} G_o^\phi(f) &= F_{eee}^\phi - F_{oee}^\phi e_1 - F_{eoe}^\phi e_2 + F_{ooe}^\phi e_3 - F_{eoo}^\phi e_4 \\ &\quad + F_{ooo}^\phi e_5 + F_{eoo}^\phi e_6 - F_{ooo}^\phi e_7, \end{aligned}$$

and the coefficients are defined by

$$\begin{aligned} F_{eee}^\phi(w, u) &= \int_{\mathbb{R}^3} h_{eee}(x, u) \cos(2\pi w_1 x_1) \cos(2\pi w_2 x_2) \cos(2\pi w_3 x_3) dx, \\ F_{oee}^\phi(w, u) &= \int_{\mathbb{R}^3} h_{oee}(x, u) \sin(2\pi w_1 x_1) \cos(2\pi w_2 x_2) \cos(2\pi w_3 x_3) dx, \\ F_{eoe}^\phi(w, u) &= \int_{\mathbb{R}^3} h_{eoe}(x, u) \cos(2\pi w_1 x_1) \sin(2\pi w_2 x_2) \cos(2\pi w_3 x_3) dx, \end{aligned}$$

$$G_o^{M_{e,1}^{w'} \phi}(M_{e,1}^{w_0} f)(w, u) = e^{e_1 2\pi u_1 w'} \cdot G_o^\phi f(w_{s,1}^1, u), \quad (33)$$

$$G_o^{M_{e,2}^{w'} \phi}(M_{e,2}^{w_0} f)(w, u) = \frac{1}{2} e^{e_2 2\pi u_2 w'} (G_o^\phi f(w_{s,7}^2, u) + G_o^\phi f(w_{s,1}^2, u) + G_o^\phi f(w_{e,3}^2, u) - G_o^\phi f(w_{e,4}^2, u)), \quad (34)$$

$$G_o^{M_{e,3}^{w'} \phi}(M_{e,3}^{w_0} f)(w, u) = \frac{1}{2} e^{e_4 2\pi u_3 w'} (G_o^\phi f(w_{s,4}^3, u) + G_o^\phi f(w_{s,2}^3, u) + G_o^\phi f(w_{e,3}^3, u) - G_o^\phi f(w_{e,4}^3, u)), \quad (35)$$

where $w_{e,3}^2 = (-w_1, w_2 + w_0 + w', w_3)$, $w_{e,3}^3 = (-w_1, -w_2, w_3 + w_0 + w')$, $w_{e,4}^2 = (-w_1, w_2 - w_0 - w', w_3)$, $w_{e,4}^3 = (-w_1, -w_2, w_3 - w_0 - w')$.

$$G_o^{(M_{e,1}^{w'} T_{a'} \phi)}(M_{e,1}^{w_0} T_a f)(w, u) = e^{e_1 2\pi u_1 w'} [\cos(2\pi \tilde{w}'_1 a) G_o^\phi f(w_{s,1}^1, z_a) - \sin(2\pi \tilde{w}'_1 a) G_o^\phi f(w_{s,2}^1, z_a) e_1], \quad (36)$$

$$\begin{aligned} & G_o^{(M_{e,2}^{w'} T_{a'} \phi)}(M_{e,2}^{w_0} T_a f)(w, u) \\ &= \frac{e^{e_2 2\pi u_2 w'}}{2} [\cos(2\pi b \tilde{w}_2) (G_o^\phi f(w_{s,7}^2, z_b) - G_o^\phi f(w_{e,4}^2, z_b)) + \cos(2\pi b \tilde{w}'_2) (G_o^\phi f(w_{s,1}^2, z_b) + G_o^\phi f(w_{e,3}^2, z_b)) \\ &\quad - \sin(2\pi b \tilde{w}_2) (G_o^\phi f(w_{s,6}^2, z_b) - G_o^\phi f(w_{e,4}^2, z_b)) e_2 - \sin(2\pi b \tilde{w}'_2) (G_o^\phi f(w_{s,2}^2, z_b) + G_o^\phi f(w_{e,3}^2, z_b)) e_2], \end{aligned} \quad (37)$$

$$\begin{aligned} & G_o^{(M_{e,3}^{w'} T_{c'} \phi)}(M_{e,3}^{w_0} T_c f)(w, u) = \frac{e^{e_4 2\pi u_3 w'}}{2} [(G_o^\phi f(w_{s,4}^3, u) - G_o^\phi f(w_{e,4}^3, u)) \cdot e^{-e_4 2\pi c \tilde{w}_3} \\ &\quad + (G_o^\phi f(w_{s,2}^3, u) + G_o^\phi f(w_{e,3}^3, u)) \cdot e^{-e_4 2\pi c \tilde{w}'_3}], \end{aligned} \quad (38)$$

where $\tilde{w}_2 = w_2 - w_0 - w'$, $\tilde{w}_3 = w_3 - w_0 - w'$, $\tilde{w}'_i = w_i + w_0 + w'$ ($i = 1, 2, 3$), $\tilde{w}_{e,3}^2 = (-w_1, w_2 + w_0 + w', -w_3)$, $\tilde{w}_{e,4}^2 = (-w_1, w_2 - w_0 - w', -w_3)$.

$$\begin{aligned}
F_{oe}^\phi(w, u) &= \int_{\mathbb{R}^3} h_{oe}(x, u) \sin(2\pi w_1 x_1) \sin(2\pi w_2 x_2) \cos(2\pi w_3 x_3) dx, \\
F_{eo}^\phi(w, u) &= \int_{\mathbb{R}^3} h_{eo}(x, u) \cos(2\pi w_1 x_1) \cos(2\pi w_2 x_2) \sin(2\pi w_3 x_3) dx, \\
F_{oeo}^\phi(w, u) &= \int_{\mathbb{R}^3} h_{oeo}(x, u) \sin(2\pi w_1 x_1) \cos(2\pi w_2 x_2) \sin(2\pi w_3 x_3) dx, \\
F_{eoo}^\phi(w, u) &= \int_{\mathbb{R}^3} h_{eoo}(x, u) \cos(2\pi w_1 x_1) \sin(2\pi w_2 x_2) \sin(2\pi w_3 x_3) dx, \\
F_{ooo}^\phi(w, u) &= \int_{\mathbb{R}^3} h_{ooo}(x, u) \sin(2\pi w_1 x_1) \sin(2\pi w_2 x_2) \sin(2\pi w_3 x_3) dx,
\end{aligned}$$

where $h(x, u) = f(x)\phi(x - u)$ is a real-valued function and it can be expressed as a union of eight terms:

$$\begin{aligned}
h(x, u) &= h_{eee}(x, u) + h_{e eo}(x, u) + h_{e oe}(x, u) + h_{e oo}(x, u) \\
&\quad + h_{oe e}(x, u) + h_{oe o}(x, u) + h_{oo e}(x, u) + h_{ooo}(x, u),
\end{aligned} \tag{44}$$

where $h_{\epsilon\epsilon\eta}(x, u)$, $\epsilon, \eta \in \{e, o\}$, are eight terms of different parity with relation to x_1, x_2 and x_3 . We use the subscripts to define even parity (e) and odd parity (o). They indicate that a function is either even (e) or odd (o) with respect to an appropriate variable, such as $h_{oe e}(x, u)$ is odd with respect to x_1 and even with respect to x_2 and x_3 .

As the classic convolution operation, we obtain the following important theorem:

Theorem 10 (Convolution theorem): Let $f, g \in L^2(\mathbb{R}^3)$ be two real-valued functions and $\phi, \varphi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be two real-valued window functions. Then we have

$$\begin{aligned}
G_o^{\phi*\varphi}(f * g)(w, u) &= \int_{\mathbb{R}^3} G_o^\varphi g(w, m) F_{eee}^\phi(w, u - m) \\
&\quad - G_o^\varphi g(t, m) F_{oe e}^\phi(w, u - m) e_1 \\
&\quad - G_o^\varphi g(s, m) F_{oe o}^\phi(w, u - m) e_2 \\
&\quad + G_o^\varphi g(t, m) F_{oo e}^\phi(w, u - m) e_3 \\
&\quad - G_o^\varphi g(w, m) F_{e eo}^\phi(w, u - m) e_4 \\
&\quad + G_o^\varphi g(s, m) F_{e oo}^\phi(w, u - m) e_5 \\
&\quad + G_o^\varphi g(t', m) F_{e oo}^\phi(w, u - m) e_6 \\
&\quad - G_o^\varphi g(t', m) F_{ooo}^\phi(w, u - m) e_7 dm,
\end{aligned} \tag{45}$$

where $t = (w_1, -w_2, -w_3) \in \mathbb{R}^3$, $s = (w_1, w_2, -w_3) \in \mathbb{R}^3$, $t' = (-w_1, w_2, -w_3) \in \mathbb{R}^3$.

Proof: According to the definition of the OSTFT and the classic convolution operation, we obtain

$$\begin{aligned}
G_o^{\phi*\varphi}(f * g)(w, u) &= \int_{\mathbb{R}^3} \left(\int_{\mathbb{R}^3} f(y) g(x - y) dy \right) \cdot \left(\int_{\mathbb{R}^3} \overline{\phi(z)} \cdot \overline{\varphi(x - u - z)} dz \right) \\
&\quad \cdot e^{-e_1 2\pi w_1 x_1} e^{-e_2 2\pi w_2 x_2} e^{-e_4 2\pi w_3 x_3} dx.
\end{aligned} \tag{46}$$

Let $x - y = q = (q_1, q_2, q_3) \in \mathbb{R}^3$, $m = u + z - y = (m_1, m_2, m_3) \in \mathbb{R}^3$, then

$$\begin{aligned}
G_o^{\phi*\varphi}(f * g)(w, u) &= \int_{\mathbb{R}^9} f(y) g(q) \overline{\phi(y - (u - m))} \cdot \overline{\varphi(q - m)} e^{-e_1 2\pi w_1 (q_1 + y_1)} \\
&\quad \cdot e^{-e_2 2\pi w_2 (q_2 + y_2)} \cdot e^{-e_4 2\pi w_3 (q_3 + y_3)} dq dy dm \\
&= \int_{\mathbb{R}^6} f(y) \overline{\phi(y - (u - m))} \left(\int_{\mathbb{R}^3} g(q) \overline{\varphi(q - m)} \right. \\
&\quad \cdot e^{-e_1 2\pi w_1 (q_1 + y_1)} e^{-e_2 2\pi w_2 (q_2 + y_2)} \\
&\quad \cdot e^{-e_4 2\pi w_3 (q_3 + y_3)} dq \left. \right) dy dm.
\end{aligned} \tag{47}$$

For every $\gamma_1, \gamma_2, \gamma_3 \in \mathbb{R}$, from the Table I, we have

$$\begin{aligned}
&((e^{-e_1 \gamma_1}) \cdot (e^{-e_2 \gamma_2})) \cdot (e^{-e_4 \gamma_3}) \\
&= ((e^{-e_1 \gamma_1} \cdot e^{-e_2 \gamma_2}) \cdot e^{-e_4 \gamma_3}), \\
&((e^{-e_1 \gamma_1} \cdot e_1) \cdot (e^{-e_2 \gamma_2})) \cdot (e^{-e_4 \gamma_3}) \\
&= ((e^{-e_1 \gamma_1} \cdot e^{e_2 \gamma_2}) \cdot e^{e_4 \gamma_3}) \cdot e_1, \\
&((e^{-e_1 \gamma_1}) \cdot (e^{-e_2 \gamma_2} \cdot e_2)) \cdot (e^{-e_4 \gamma_3}) \\
&= ((e^{-e_1 \gamma_1} \cdot e^{-e_2 \gamma_2}) \cdot e^{e_4 \gamma_3}) \cdot e_2, \\
&((e^{-e_1 \gamma_1} \cdot e_1) \cdot (e^{-e_2 \gamma_2} \cdot e_2)) \cdot (e^{-e_4 \gamma_3}) \\
&= ((e^{-e_1 \gamma_1} \cdot e^{e_2 \gamma_2}) \cdot e^{e_4 \gamma_3}) \cdot e_3, \\
&((e^{-e_1 \gamma_1}) \cdot (e^{-e_2 \gamma_2})) \cdot (e^{-e_4 \gamma_3} \cdot e_4) \\
&= ((e^{-e_1 \gamma_1} \cdot e^{-e_2 \gamma_2}) \cdot e^{-e_4 \gamma_3}) \cdot e_4, \\
&((e^{-e_1 \gamma_1} \cdot e_1) \cdot (e^{-e_2 \gamma_2})) \cdot (e^{-e_4 \gamma_3} \cdot e_4) \\
&= ((e^{-e_1 \gamma_1} \cdot e^{-e_2 \gamma_2}) \cdot e^{e_4 \gamma_3}) \cdot e_5, \\
&((e^{-e_1 \gamma_1}) \cdot (e^{-e_2 \gamma_2} \cdot e_2)) \cdot (e^{-e_4 \gamma_3} \cdot e_4) \\
&= ((e^{e_1 \gamma_1} \cdot e^{-e_2 \gamma_2}) \cdot e^{e_4 \gamma_3}) \cdot e_6, \\
&((e^{-e_1 \gamma_1} \cdot e_1) \cdot (e^{-e_2 \gamma_2} \cdot e_2)) \cdot (e^{-e_4 \gamma_3} \cdot e_4) \\
&= ((e^{e_1 \gamma_1} \cdot e^{-e_2 \gamma_2}) \cdot e^{e_4 \gamma_3}) \cdot e_7.
\end{aligned}$$

Furthermore, we can rewrite the transformation kernel in the following way

$$\begin{aligned}
&e^{-e_1 2\pi w_1 (q_1 + y_1)} e^{-e_2 2\pi w_2 (q_2 + y_2)} e^{-e_4 2\pi w_3 (q_3 + y_3)} \\
&= ((e^{-e_1 \alpha_1} \cdot \cos(\beta_1)) \cdot (e^{-e_2 \alpha_2} \cdot \cos(\beta_2))) \\
&\quad \cdot (e^{-e_4 \alpha_3} \cdot \cos(\beta_3)) - ((e^{-e_1 \alpha_1} \cdot e_1 \sin(\beta_1)) \\
&\quad \cdot (e^{-e_2 \alpha_2} \cdot \cos(\beta_2))) \cdot (e^{-e_4 \alpha_3} \cdot \cos(\beta_3))
\end{aligned}$$

$$\begin{aligned}
& -((e^{-e_1\alpha_1} \cdot \cos(\beta_1)) \cdot (e^{-e_2\alpha_2} \cdot e_2 \sin(\beta_2))) \\
& \cdot (e^{-e_4\alpha_3} \cdot \cos(\beta_3)) + ((e^{-e_1\alpha_1} \cdot e_1 \sin(\beta_1)) \\
& \cdot (e^{-e_2\alpha_2} \cdot e_2 \sin(\beta_2))) \cdot (e^{-e_4\alpha_3} \cdot \cos(\beta_3)) \\
& -((e^{-e_1\alpha_1} \cdot \cos(\beta_1)) \cdot (e^{-e_2\alpha_2} \cdot \cos(\beta_2))) \\
& \cdot (e^{-e_4\alpha_3} \cdot e_4 \sin(\beta_3)) + ((e^{-e_1\alpha_1} \cdot e_1 \sin(\beta_1)) \\
& \cdot (e^{-e_2\alpha_2} \cdot \cos(\beta_2))) \cdot (e^{-e_4\alpha_3} \cdot e_4 \sin(\beta_3)) \\
& + ((e^{-e_1\alpha_1} \cdot \cos(\beta_1)) \cdot (e^{-e_2\alpha_2} \cdot e_2 \sin(\beta_2))) \\
& \cdot (e^{-e_4\alpha_3} \cdot e_4 \sin(\beta_3)) - ((e^{-e_1\alpha_1} \cdot e_1 \sin(\beta_1)) \\
& \cdot (e^{-e_2\alpha_2} \cdot e_2 \sin(\beta_2))) \cdot (e^{-e_4\alpha_3} \cdot e_4 \sin(\beta_3)).
\end{aligned}$$

where $\alpha_i = 2\pi w_i q_i$, $\beta_i = 2\pi w_i y_i$, $i = 1, 2, 3$. Hence, the (47) can be written the (45), where $t = (w_1, -w_2, -w_3) \in \mathbb{R}^3$, $s = (w_1, w_2, -w_3) \in \mathbb{R}^3$, $t' = (-w_1, w_2, -w_3) \in \mathbb{R}^3$.

V. PITT'S INEQUALITY AND LIEB'S INEQUALITY

In [33], Lian proved a sharp Pitt's inequality of the OFT, as follows:

Lemma 5: For $f \in L^2(\mathbb{R}^3)$, $0 \leq \rho \leq 3$, then

$$\int_{\mathbb{R}^3} |w|^{-\rho} |F_o(f)(w)|^2 dw \leq C_\rho \int_{\mathbb{R}^3} |x|^\rho |f(x)|^2 dx,$$

where $C_\rho = \pi^\rho \left(\frac{\Gamma(\frac{3-\rho}{4})}{\Gamma(\frac{3+\rho}{4})} \right)^2$.

Based on Pitt's inequality for the OFT, we obtain Pitt's inequality for the OSTFT.

Theorem 11 (Pitt's inequality): Let $f \in L^2(\mathbb{R}^3)$ be a real-valued functions and $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a real-valued window function. Then we have

$$\begin{aligned}
& \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} |w|^{-\rho} |G_o^\phi(f)(w, u)|^2 dw du \\
& \leq C_\rho \|\phi\|_{L^2(\mathbb{R}^3)}^2 \int_{\mathbb{R}^3} |x|^\rho |f(x)|^2 dx,
\end{aligned} \quad (48)$$

where $C_\rho = \pi^\rho \left(\frac{\Gamma(\frac{3-\rho}{4})}{\Gamma(\frac{3+\rho}{4})} \right)^2$.

Proof: Let $h(x, u) = f(x)\phi(x - u)$, according to Lemma 5 and (6), we have

$$\begin{aligned}
\int_{\mathbb{R}^3} |w|^{-\rho} |G_o^\phi(f)(w, u)|^2 dw &= \int_{\mathbb{R}^3} |w|^{-\rho} |F_o(h)(w)|^2 dw \\
&\leq C_\rho \int_{\mathbb{R}^3} |x|^\rho |h(x)|^2 dx \\
&= C_\rho \int_{\mathbb{R}^3} |x|^\rho |f(x)\phi(x - u)|^2 dx.
\end{aligned} \quad (49)$$

Integrating both sides with respect to du in (49). We obtain the result.

$$\begin{aligned}
& \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} |w|^{-\rho} |G_o^\phi(f)(w, u)|^2 dw du \\
& \leq C_\rho \|\phi\|_{L^2(\mathbb{R}^3)}^2 \int_{\mathbb{R}^3} |x|^\rho |f(x)|^2 dx,
\end{aligned}$$

Let us give a short and simple proof of Lieb's inequality of the OSTFT by considering the relationship between the OSTFT and the OFT.

Theorem 12 (Lieb's inequality): Let $\phi \in L^2(\mathbb{R}^3) \setminus \{0\}$ be a real-valued window function. For every real-valued function $f \in L^2(\mathbb{R}^3)$ and $p \geq 2$, we have

$$\|G_o^\phi(f)\|_{L^p(\mathbb{R}^3)} \leq A_s^{\frac{6}{p}} A_{t'}^{\frac{3}{p}} \|f\|_{L^2(\mathbb{R}^3)} \|\phi\|_{L^2(\mathbb{R}^3)}, \quad (50)$$

where $A_s = \left(\frac{s^{\frac{1}{s}}}{s^{\frac{1}{s'}}} \right)^{\frac{1}{2}}$, $A_{t'} = \left(\frac{t'^{\frac{1}{t'}}}{t^{\frac{1}{t}}} \right)^{\frac{1}{2}}$.

Proof: According to the Hausdorff-Young theorem of the OFT [33], we obtain

$$\begin{aligned}
\left(\int_{\mathbb{R}^3} |G_o^\phi(f)(w, u)|^p dw \right)^{\frac{1}{p}} &= \left(\int_{\mathbb{R}^3} |F_o(h)(w)|^p dw \right)^{\frac{1}{p}} \\
&\leq \left(\int_{\mathbb{R}^3} |h(x)|^q dx \right)^{\frac{1}{q}} \\
&= \left(|f|^q * |\tilde{\phi}|^q(u) \right)^{\frac{1}{q}},
\end{aligned} \quad (51)$$

where $h(x, u) = f(x)\overline{\phi(x - u)}$ and $\tilde{\phi}(x) = \phi(-x)$.

Hence

$$\begin{aligned}
\|G_o^\phi(f)\|_p &= \left(\int_{\mathbb{R}^3} \left(\int_{\mathbb{R}^3} |G_o^\phi(f)(w, u)|^p dw \right) du \right)^{\frac{1}{p}} \\
&\leq \left(\int_{\mathbb{R}^3} \left(|f|^q * |\tilde{\phi}|^q(u) \right)^{\frac{p}{q}} du \right)^{\frac{1}{p}} \\
&= \| |f|^q * |\tilde{\phi}|^q \|_{L^{\frac{p}{q}}(\mathbb{R}^3)}^{\frac{1}{p}}.
\end{aligned} \quad (52)$$

If $s = \frac{2}{q}$, $t = \frac{p}{q}$ and $\frac{1}{s} + \frac{1}{s'} = 1$, $\frac{1}{t} + \frac{1}{t'} = 1$, then $\frac{1}{s} + \frac{1}{s'} = 1 + \frac{1}{t}$ and $|f|^q, |\tilde{\phi}|^q \in L^2(\mathbb{R}^3)$.

By Young inequality, we obtain

$$\| |f|^q * |\tilde{\phi}|^q \|_{L^t(\mathbb{R}^3)} \leq A_s^6 A_{t'}^3 \| |f|^q \|_{L^s(\mathbb{R}^3)} \| |\tilde{\phi}|^q \|_{L^s(\mathbb{R}^3)}, \quad (53)$$

where $A_s = \left(\frac{s^{\frac{1}{s}}}{s^{\frac{1}{s'}}} \right)^{\frac{1}{2}}$, $A_{t'} = \left(\frac{t'^{\frac{1}{t'}}}{t^{\frac{1}{t}}} \right)^{\frac{1}{2}}$.

However

$$\| |f|^q \|_{L^s(\mathbb{R}^3)} = \left(\int_{\mathbb{R}^3} |f(x)|^{q \cdot \frac{2}{q}} dx \right)^{\frac{q}{2}} = \|f\|_{L^2(\mathbb{R}^3)}^q, \quad (54)$$

and

$$\| |\tilde{\phi}|^q \|_{L^s(\mathbb{R}^3)} = \left(\int_{\mathbb{R}^3} |\phi(x - u)|^{q \cdot \frac{2}{q}} dx \right)^{\frac{q}{2}} = \|\phi\|_{L^2(\mathbb{R}^3)}^q. \quad (55)$$

Hence

$$\| |f|^q * |\tilde{\phi}|^q \|_{L^t(\mathbb{R}^3)}^{\frac{1}{q}} \leq \left(A_s^6 A_{t'}^3 \|f\|_{L^2(\mathbb{R}^3)}^q \|\phi\|_{L^2(\mathbb{R}^3)}^q \right)^{\frac{1}{q}}. \quad (56)$$

We have

$$\begin{aligned}
\|G_o^\phi(f)\|_{L^p(\mathbb{R}^3)} &\leq \left(A_s^6 A_{t'}^3 \|f\|_{L^2(\mathbb{R}^3)}^q \|\phi\|_{L^2(\mathbb{R}^3)}^q \right)^{\frac{1}{q}} \\
&= A_s^{\frac{6}{q}} A_{t'}^{\frac{3}{q}} \|f\|_{L^2(\mathbb{R}^3)} \|\phi\|_{L^2(\mathbb{R}^3)}.
\end{aligned} \quad (57)$$

Remark 1: If $\|\phi\|_{L^2(\mathbb{R}^3)} = 1$, then we have

$$\|G_o^\phi(f)\|_{L^p(\mathbb{R}^3)} \leq A_s^{\frac{6}{p}} A_{t'}^{\frac{3}{p}} \|f\|_{L^2(\mathbb{R}^3)}.$$

VI. LOGARITHMIC UNCERTAINTY PRINCIPLE

As a solid theoretical basis, the research tendency of TFA is to enhance the TF resolution and energy concentration. The uncertainty principle makes a tradeoff between temporal and spectral resolutions unavoidable. In this section, we used the logarithmic uncertainty principle of the OFT [33] to obtain the logarithmic uncertainty principle of the OSTFT.

Lemma 6: For $f \in S(\mathbb{R}^3)$,

$$\begin{aligned} & \int_{\mathbb{R}^3} \ln |x| |f(x)|^2 dx + \int_{\mathbb{R}^3} \ln |w| |F_o(f)(w)|^2 dw \\ & \geq D \int_{\mathbb{R}^3} |f(x)|^2 dx, \end{aligned} \quad (58)$$

where $D = \psi(\frac{3}{4}) + \ln(2)$, $\psi = \frac{d}{dt}(\ln \Gamma(t))$ and $\Gamma(t)$ is the Gamma function.

Theorem 13: Let $f, \phi \in S(\mathbb{R}^3)$, then we have the following inequality

$$\begin{aligned} & \|\phi\|_{L^2(\mathbb{R}^3)}^2 \int_{\mathbb{R}^3} \ln |x| |f(x)|^2 dx \\ & + \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \ln |w| |G_o^\phi(f)(w, u)|^2 dw du \\ & \geq D \|f\|_{L^2(\mathbb{R}^3)}^2 \|\phi\|_{L^2(\mathbb{R}^3)}^2, \end{aligned} \quad (59)$$

where $D = \psi(\frac{3}{4}) + \ln(2)$, $\psi = \frac{d}{dt}(\ln \Gamma(t))$ and $\Gamma(t)$ is the Gamma function.

Proof: Since $f, \phi \in S(\mathbb{R}^3)$, this implies that $h(x, u) = f(x)\phi(x-u) \in S(\mathbb{R}^3)$. Therefore, we can replace f by h on both sides of (58) and obtain

$$\begin{aligned} & \int_{\mathbb{R}^3} \ln |x| |h(x, u)|^2 dx + \int_{\mathbb{R}^3} \ln |w| |F_o(h)(w)|^2 dw \\ & \geq D \int_{\mathbb{R}^3} |h(x, u)|^2 dx, \end{aligned} \quad (60)$$

integration both sides of this equation with respect to du yields

$$\begin{aligned} & \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \ln |x| |h(x, u)|^2 dx du + \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \ln |w| |F_o(h)(w)|^2 dw du \\ & \geq D \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} |h(x, u)|^2 dx du. \end{aligned} \quad (61)$$

According to (6), we obtain

$$\begin{aligned} & \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \ln |x| |f(x)|^2 |\phi(x-u)|^2 dx du \\ & + \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \ln |w| |G_o^\phi(f)(w, u)|^2 dw du \\ & \geq D \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} |f(x)|^2 |\phi(x-u)|^2 dx du. \end{aligned} \quad (62)$$

Hence, we obtain the result.

Remark 2: When the window function is normalized, then (59) implies

$$\begin{aligned} & \int_{\mathbb{R}^3} \ln |x| |f(x)|^2 dx + \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \ln |w| |G_o^\phi(f)(w, u)|^2 dw du \\ & \geq D \|f\|_{L^2(\mathbb{R}^3)}^2. \end{aligned} \quad (63)$$

VII. APPLICATIONS

When analyzing signals and systems with LTV properties, some transforms are used as a joint time-frequency representation (TFR) of the signal [1], [21], [38], [39]. In this section, the newly proposed OSTFT is applied to form an LTV system. We may regard the OSTFT as an LTV filterbank operation. Therefore, the LTV spectrum obtained by the OSTFT can also be interpreted as the output of an LTV filterbank.

Definition 4: Let a LTV system with impulse response $r(\cdot, \cdot, \cdot)$. Then, the output $o(\cdot, \cdot)$ of the LTV system is defined by

$$o(w, u) = \int_{\mathbb{R}^3} f(x) r(w, u, u-x) dx, \quad (64)$$

where $f(x)$ is the input signal.

The transfer function $R_o(\cdot, \cdot)$ of the impulse response $r(\cdot, \cdot, \cdot)$ is

$$R_o(w, u) = \int_{\mathbb{R}^3} r(w, u, t) e^{-e_1 2\pi t_1 w_1} e^{-e_2 2\pi t_2 w_2} e^{-e_4 2\pi t_3 w_3} dt, \quad (65)$$

where $t = (t_1, t_2, t_3) \in \mathbb{R}^3$.

Based on the above definition, we develop the following theorem which links the stationary concepts of the OSTFT to those of LTV filtering.

Theorem 14: The LTV system with impulse response $r_1(\cdot, \cdot, \cdot)$ of the filter can be defined by

$$\begin{aligned} & r_1(w, u, t) \\ & = \overline{\phi(-t)} e^{-e_1 2\pi(u_1-t_1)w_1} e^{-e_2 2\pi(u_2-t_2)w_2} e^{-e_4 2\pi(u_3-t_3)w_3}. \end{aligned} \quad (66)$$

If the input to this system is the signal $f(x)$, the output signal of the system is equal to the OSTFT of the input signal.

Proof: According to Definition 4, we obtain

$$\begin{aligned} & o_1(w, u) = \int_{\mathbb{R}^3} f(x) r_1(w, u, u-x) dx \\ & = G_o^\phi(f)(w, u), \end{aligned} \quad (67)$$

where $\phi(\cdot)$ is the window function.

To obtain some insight into the TV filters corresponding to the above contents, let us give an example.

Example 1: The LTV impulse response is defined by (66). The window function is the rectangular window function defined by

$$\phi(x) := \begin{cases} 1, & -\frac{1}{2} \leq x_1 \leq \frac{1}{2}, -\frac{1}{2} \leq x_2 \leq \frac{1}{2} \\ & \text{and } -\frac{1}{2} \leq x_3 \leq \frac{1}{2}, \\ 0, & \text{otherwise.} \end{cases} \quad (68)$$

The output $o_1(\cdot, \cdot, \cdot)$ of the input signal $f(x)$ is found. The input signal is defined by

$$f(x) := \begin{cases} e^{x_1+x_2+x_3}, & -\infty < x_1 < 0, -\infty < x_2 < 0 \\ & \text{and } -\infty < x_3 < 0, \\ 0, & \text{otherwise.} \end{cases} \quad (69)$$

According to Theorem 14, we obtain

$$\begin{aligned} o_1(w, u) &= \int_{\mathbb{R}^3} f(x) \overline{\phi(x-u)} e^{-e_1 2\pi x_1 w_1} e^{-e_2 2\pi x_2 w_2} e^{-e_4 2\pi x_3 w_3} dx \\ &= \int_{u_1-\frac{1}{2}}^{v_1} \int_{u_2-\frac{1}{2}}^{v_2} \int_{u_3-\frac{1}{2}}^{v_3} e^{x_1+x_2+x_3} e^{-e_1 2\pi x_1 w_1} e^{-e_2 2\pi x_2 w_2} \\ &\quad \cdot e^{-e_4 2\pi x_3 w_3} dx_1 dx_2 dx_3 \\ &= \int_{u_1-\frac{1}{2}}^{v_1} e^{x_1 \beta_1} dx_1 \int_{u_2-\frac{1}{2}}^{v_2} e^{x_2 \beta_2} dx_2 \int_{u_3-\frac{1}{2}}^{v_3} e^{x_3 \beta_3} dx_3 \\ &= \frac{e^{v_1 \beta_1} - e^{(u_1-\frac{1}{2}) \beta_1}}{\beta_1} \cdot \frac{e^{v_2 \beta_2} - e^{(u_2-\frac{1}{2}) \beta_2}}{\beta_2} \\ &\quad \cdot \frac{e^{v_3 \beta_3} - e^{(u_3-\frac{1}{2}) \beta_3}}{\beta_3}, \end{aligned} \quad (70)$$

where $v_1 = \min(0, \frac{1}{2} + u_1)$, $v_2 = \min(0, \frac{1}{2} + u_2)$ and $v_3 = \min(0, \frac{1}{2} + u_3)$; $\beta_1 = 1 - e_1 2\pi w_1$, $\beta_2 = 1 - e_2 2\pi w_2$ and $\beta_3 = 1 - e_3 2\pi w_3$.

It should be noted that the multiplication in the above formula is done from left to right.

In addition to the analysis of the LTV system of the OSTFT, there are also many other possible applications for the OSTFT. As a new TFA tool, a potential application area of the OSTFT may be color image processing. There is a classical method for color image processing, in which hypercomplex FT treats the multichannel signals as an algebraic whole without losing the spectral relations. However, the hypercomplex FT cannot satisfactorily extract local information well due to its global kernel function. The OSTFT can handle this problem well by using a sliding window. Another potential application of the OSTFT is to reconstruct each monocomponent mode from a multicomponent signal [6]. Furthermore, an issue is to investigate the relationships between the OSTFT and some other TFA distributions, such as the wavelet transform [40], synchrosqueezing transform [8], general linear chirplet transform [7], Wigner distribution (WD) and ambiguity function (AF) [24]. It is very propitious to derive novel nonstationary signal processing techniques.

VIII. CONCLUSION

In the current work, we present a novel concept of the OSTFT by generalizing the QSTFT to the octonion algebra setting and obtain the relations between the OSTFT and the OFT. Then, the various properties of the proposed OSTFT are explored, such as linearity, inversion formulas, shift and orthogonality relation. Based on the classic Fourier convolution operation, the convolution theorem for the OSTFT is derived. Furthermore, Pitt's

inequality and Lieb's inequality associated with the OSTFT are obtained. From the logarithmic uncertainty principle of the OFT, the logarithmic uncertainty principle associated with the OSTFT is investigated. Finally, an example and several potential applications of the OSTFT are presented.

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